

Master's Thesis of Data Science

Action-Centric Query Understanding in Referring Image Segmentation

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Abstract

Referring Image Segmentation (RIS) requires precise multimodal understanding, particularly when interpreting action-centric expressions. Existing models often fail to effectively utilize verbs and their associated semantics, leading to diminished performance on motion-oriented queries. To address this, we propose the Verb-Centric Multimodal Radial Contrastive Loss (MRaCL), a novel training objective that directly manipulates angular representations by converting angular distances in radians into logits. This approach allows for more effective gradient flow and better separation of embeddings in densely clustered multimodal spaces, overcoming limitations of conventional cosine similarity. By leveraging verb phrases extracted with large language models, MRaCL constructs hard positives that emphasize action-oriented semantics, aligning motion-centric language with corresponding visual cues. Our method significantly enhances RIS performance on challenging motion-centric benchmarks, advancing the model’s ability to comprehend dynamic and nuanced language-visual relationships.

Keywords: Referring Image Segmentation, Action-Oriented semantics, Multimodal Radial Contrastive Loss, Cross-Modal Alignment

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Chapter 1

Introduction

Referring Image Segmentation (RIS) is a multimodal task that involves identifying the target referent within an image based on a given language expression. Its complexity arises from resolving visual ambiguities and linguistic intricacies, which vary across instances. Modern RIS datasets have evolved to incorporate increasingly complex linguistic and visual elements, with richer textual queries providing nuanced cues for object localization. Notably, as shown in Table 1.1, G-Ref [36] exhibits a longer average length of referring expressions compared to RefCOCO(+) [51] and features action verbs in 49% of queries, highlighting its emphasis on motion and action-based cues. These motion-centric expressions frequently provide critical context for identifying referents, underscoring the importance of understanding how well existing models grasp such cues. The question remains: **Can RIS models effectively interpret action-oriented expressions in queries?** More importantly, can these models resolve ambiguities stemming from these expressions in the image domain?



Figure 1.1: **Comparing model performance on static attributes versus motion-centric queries.** In a.1, the static query “baby in rainbow sweater” focuses on appearance, while a.2, the motion-centric query “baby being held”, emphasizes action. Similarly, b.1 highlights static attributes like “orange jacket and white helmet”, whereas b.2, “bending over and touching his boots”, tests the model’s ability to interpret motion.

Table 1.1: Statistics of representative RIS Datasets.

Dataset	RefCOCO	RefCOCO+	G-Ref
# Images	19,994	19,992	26,711
# Ref. Exp.	142,209	141,564	85,474
Query length	3.61	3.53	8.43
Obj. per query	1.76	1.67	3.03
Action* Freq	13.63%	14.67%	49.26%
Static* Freq	50.24%	49.52%	49.95%

Table 1.2: mIoU & oIoU on motion-centric and static attribute-centric samples from G-Ref UMD test set.

Models	mIoU		oIoU	
	Static	Motion	Static	Motion
LAVT [48]	72.59	55.05	70.63	49.78
CRIS [44]	70.36	52.82	65.28	48.37
CGFormer [40]	75.86	58.22	74.95	52.27

Existing RIS models often overlook the nuanced motion and relational dynamics conveyed by verbs, despite their potential as a “free” supervision source. Current state-of-the-art models trained on existing dataset settings tend to heavily focus on object names while disregarding verbs or actions. For instance, motion-centric expressions, which often serve as key discriminative cues in queries, are underutilized. To investigate this, we manually curated a motion-centric and static attribute-based split (500 samples each) using samples from the G-Ref [36] UMD test split. This allows us to isolate and analyze the impact of action-based verbs versus static attributes in model performance. This breakdown is essential for distinguishing whether the difficulty arises from motion-



Figure 1.2: **Sentence decomposition results highlighting CGFormer’s reliance on noun phrases.** In the query “Taller child holding ski-sticks upright”, CGFormer focuses on the noun phrase “taller child”. Similarly, in “a cow eating grass”, the model emphasizes “cow” and “grass”, overlooking the action.

centric expressions themselves or from the added complexity of attribute-laden descriptions. For example, as shown in Fig. 1.1, both a motion-centric query like “bending over and touching” and an attribute-based query like “orange jacket and white helmet” may refer to the same target object, yet they challenge the model in fundamentally different ways.

Table 1.2 shows a significant performance gap between motion-centric and attribute-centric samples when evaluated with current RIS models. The results highlight that these models perform well in identifying objects when provided with explicit noun phrases or static attributes but struggle to interpret motion-centric words effectively. This deficiency is particularly pronounced when the query requires distinguishing between multiple objects of the same category based on their motion or relational dynamics.

Further qualitative analysis through sentence decomposition corroborates

these findings. As shown in Fig. 1.2, when queries are broken down into their key components, CGFormer [40] demonstrates a strong reliance on noun phrases while exhibiting limited capacity to infer targets based on action-related cues. For instance, in the query “Taller child holding ski-sticks upright,” models often ignore the verb “holding” and instead focus exclusively on the noun “child,” leading to incorrect predictions when nouns alone do not uniquely identify the target. This highlights the models’ inability to integrate motion-centric information, even when it provides essential context.

These observations point to a critical gap in the current RIS training paradigms and dataset design. The over-reliance on explicit, information-rich noun phrases or noun phrase-based syntactic structures suggests that existing datasets may lack the linguistic diversity needed to develop robust multimodal understanding. Moreover, the absence of benchmarks specifically designed to evaluate action verb-oriented comprehension in RIS tasks further limits progress in this area. Addressing this gap requires a systematic reevaluation of how datasets are constructed and how models are trained to ensure that verbs and motion-centric expressions are adequately emphasized and effectively leveraged in RIS tasks.

Given this problem landscape, we aim to improve performance by addressing a critical yet underexplored aspect of training: enhancing the exposure of models to unseen verb structures and action(verb)-oriented expressions. We postulate that enabling models to learn the relationships between these novel expressions and their existing in-batch counterparts during training could significantly improve their capacity to comprehend the subtle and diverse patterns inherent in action-centric queries. However, creating entirely new captions and annotations to support this endeavor is prohibitively expensive, presenting a

significant practical challenge.

To overcome the limitations posed by data scarcity while achieving a compositional understanding of verbs, we propose leveraging contrastive learning in conjunction with augmentation techniques. A key focus is placed on constructing hard positives—structured examples that emphasize verb-centric understanding—while avoiding the pitfalls of overly difficult hard negatives that can degrade performance when the model lacks foundational verb knowledge. Instead, our method leverages a weakly supervised learning approach to incorporate verb phrases as hard positives, modeling the relationships between verbs and ground truth captions. This strategy enables models to better interpret verb phrases, which often represent unseen or complex sentence structures, without requiring additional datasets or extensive reannotation.

Existing contrastive losses face two key challenges: (1) the lack of a well-defined decision margin, leaving embeddings vulnerable to boundary perturbations, and (2) vanishing gradients at extreme similarity values, especially in anisotropic embedding spaces where representations converge into a narrow cone. These issues are particularly problematic in multimodal tasks like Referring Image Segmentation (RIS), where sentence-conditioned visual embeddings are tightly clustered. To address this, we propose the **Multimodal Radial Contrastive Loss (MRaCL)**, which introduces a margin penalty into angle-based similarity computation to ensure stronger alignment for positive samples and effective separation from negatives. Unlike cosine-based methods like SimCSE [14], MRaCL delivers more effective gradient flow and improved discriminative power, significantly enhancing performance on complex datasets and our novel action-centric benchmark.

Our main contributions are summarized as follows:

1. We address the challenges of action-centric queries by introducing a framework that integrates motion and verb-related semantics, improving the grounding of dynamic expressions in RIS.
2. We propose a Verb-focused Radial Contrastive embedding framework tailored for multimodal settings, which mitigates the limitations of cosine-based losses and alleviates embedding space clustering.
3. We present new motion and static evaluation splits based on RefCOCO, along with M-Bench, a benchmark designed for rigorous and quantitative assessment of motion understanding.

Chapter 2

Related Works

2.1 Referring Image Segmentation

Existing studies [16] have concentrated on encoder and decoder architectures to handle multimodal features: RNNs [27, 30, 37, 38] and Transformers [1] for text features, and CNNs [16, 30], DeeplabV3 [1, 3, 27], and DarkNet [21, 34] for visual encoding. More recently, Transformer-based backbones [21, 33, 10, 22] and multi-scale features [2, 18, 20, 49] have gained popularity for capturing detailed object masks.

Visual-linguistic fusion methods have evolved from simple concatenation [16] to more sophisticated approaches leveraging syntax [19, 20, 50]. To improve cross-modal fusion, LAVT [48] and DMMI [17] introduced early fusion strategies within vision encoder stages, integrating visual and linguistic features before feature extraction is complete. In contrast, methods like VLT [11], CRIS [44], and ReSTR [24] separately encode visual and linguistic inputs using dual encoders and then merge these features through a multi-modal transformer en-

coder or a cross-modal decoder. Similarly, CGFormer [40] organizes visual features into language-conditioned tokens to capture region-level information. Beyond these fusion approaches, methods such as CrossVLT [6] and CoupAlign [54] aim to achieve multi-level alignment by allowing multimodal interactions at each stage of their respective encoders. Additionally, VPD [55] employs a multi-scale feature map from a text-to-image diffusion model, while ReLA [29] and DMMI [17] extend RIS to support an arbitrary number of targets.

2.2 Related Datasets and Challenges

Initially, ReferIt [23], RefCOCO [51], and RefCOCO+ [51] were introduced as benchmarks for single-target Referring Image Segmentation (RIS). RefCOCO contains many positional words such as “front” or “the third from the right,” while RefCOCO+ prohibits such direct usage of positional terms. However, many examples in these benchmarks present over-simplified scenarios with short queries, often lacking sufficient ambiguities in the images.

In contrast, more recent datasets embrace complex scenes and intricate linguistic expressions. G-Ref [36] features relatively longer sentences, making the task more challenging. PhraseCut [46], built upon [25], provides examples involving multiple targets with attributes, categories, and relationships described in phrases. GRES [29] further increases the complexity by incorporating examples referencing no target or multiple objects. Similarly, Ref-ZOM, proposed by DMMI [17], is akin to gRefCOCO [29], posing challenges related to one-to-one, one-to-many, and one-to-zero referring expressions to targets in image relationships within RIS. Additionally, CGFormer [40] introduces new splits on

the RefCOCO datasets to evaluate generalization capability on objects from unseen classes.

Our method aligns with these research efforts but diverges in its focus. Unlike dataset creation or augmentation, we emphasize verb-in-action-centric understanding. Specifically, our goal is to leverage existing data to generate hard positive text examples, enhancing models’ ability to handle complex and ambiguous expressions effectively.

2.3 Deep Metric Learning Objectives

Deep Metric Learning (DML) involves mapping objects into an embedded space where similar objects are positioned closer together, and dissimilar ones are pushed farther apart. Early approaches, such as Contrastive Loss [7], N-Pair Loss [39], and Triplet Margin Loss [35], laid the groundwork for this field. Subsequently, softmax-based objectives like Center Loss [45], SphereFace [32], and CosFace [42] further advanced supervised learning methods. Angular-margin techniques, such as ArcFace [9] and AoE [28], focused on improving geometric representations, although their approaches differ. For instance, AoE mitigates issues like cosine saturation through angle-based learning in complex spaces, whereas ArcFace depends on class labels, limiting its scalability to multimodal tasks. Most of these methods are predominantly applied in single-modality domains, such as text or vision. Our proposed Multimodal Radial Contrastive Loss (MRaCL), by contrast, avoids reliance on class labels, excels in pairwise modeling for text-conditioned embeddings, and demonstrates robust discriminative capabilities in complex multimodal scenarios.

Chapter 3

MRaCL : Multimodal Radial Contrastive Learning

3.1 Preliminary

To model both positive and negative relationships between embeddings, the process begins by creating representations and forming them into positive and negative pairs. These pairs serve as input for a training objective that is fine-tuned during the learning phase.

SimCSE [14] introduced a foundational framework for learning sentence representations, influencing many recent studies [56, 52, 8, 53, 31]. In this approach, given a sentence dataset $\mathcal{D} = \{x_i\}_{i=1}^m$, positive pairs are obtained by independently encoding each sentence x_i twice using dropout-based augmentation, while negative pairs are constructed by in-batch negative sampling [5]. The learning objective is based on the loss of NT-Xent [4], which is derived from a similarity function $\text{sim}(h_i, h_j)$. Specifically, the cosine similarity between two

representations h_i and h_j is calculated as follows:

$$\text{sim}(h_i, h_j) = \frac{h_i \cdot h_j}{\|h_i\| \|h_j\|}$$

The NT-Xent loss is expressed as:

$$\mathcal{L}_{\text{NT-Xent}} = -\log \frac{e^{\text{sim}(h_i, h_i^*)/\tau}}{\sum_{j=1}^n e^{\text{sim}(h_i, h_j)/\tau}}$$

In this formulation, h_i refers to the anchor representation, h_i^* represents the corresponding positive example, and h_j ($i \neq j$) corresponds to negative examples. The parameter τ (temperature) regulates the sharpness of the softmax distribution, while n indicates the total number of samples in the batch.

3.2 Angle Based Contrastive Learning

Most existing works, including SimCSE [14], utilize a naive cosine similarity as the similarity function. Building on this, the goal is to train the model to pull representations with similar semantics closer while pushing apart those with dissimilar meanings. However, this objective has two significant limitations.

First, the issue of decision boundaries arises. While the training objective aims to pull semantically similar representations closer and push dissimilar ones farther apart, the resulting representations often lack sufficient discriminative power. Without a well-defined margin, small perturbations near the decision boundary can lead to incorrect predictions. The decision boundary is defined as the point where the exponential of pairwise similarity scores for positive and negative samples becomes equal. Introducing a margin is essential: negative

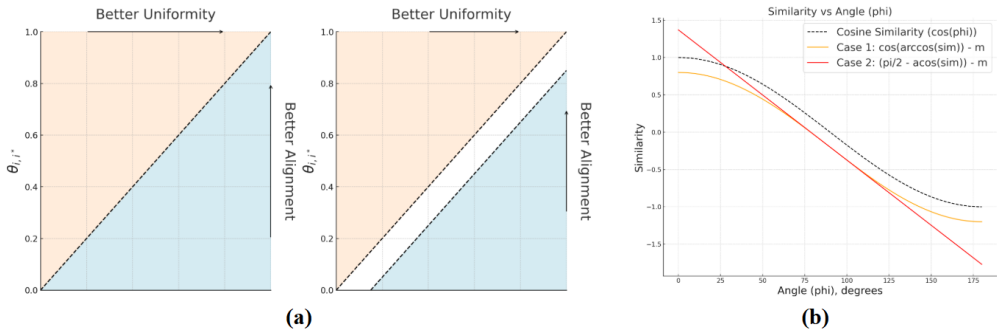


Figure 3.1: **Decision Margin and Gradient Stability in Angular Space.**

(a) shows how m ensures better separation between positive and negative pairs, enhancing robustness near the decision boundary. (b) compares cosine similarity (black dashed), cosine space margin ($\cos(\phi + m)$, orange), and angular space margin ($\pi/2 - \phi - m$, red), where $\phi = \arccos(\text{sim})$. Angular space provides more stable gradients, mitigating anisotropy and vanishing gradient issues.

samples must remain at least a margin away from positive samples in angular space to ensure robust weight assignments.

Second, cosine similarity has inherent limitations, particularly when used in contrastive loss. Cosine similarity maps the angle between two vectors to a similarity score. However, when the angle between vectors is very small (close to 0°), the cosine value approaches $+1$, and similarly, it approaches -1 for angles close to 180° . Near these extremes, gradients flatten, leading to vanishing gradients. This hinders optimization, especially when embeddings are either too closely aligned or well-separated.

This issue is exacerbated by the anisotropic nature of text encoder embeddings, which occupy only a small region of the embedding space. Pretrained language models often exhibit this anisotropy, causing sentence representations to collapse into a narrow cone [13, 12, 43, 26]. Similarly, in referring image segmentation (RIS) models, cross-modal embeddings inherit this problem. The sentence-conditioned visual representations also converge into a narrow space,

making it difficult to optimize contrastive loss effectively. When positive and negative samples have very similar angles, cosine similarity fails to provide meaningful gradients, limiting the model’s ability to learn effectively.

By contrast, angular contrastive loss mitigates this issue by ensuring that gradients flow more effectively in regions where similarity is high, allowing the model to better differentiate between positive and negative pairs in challenging cases. This approach imposes a stricter constraint, requiring that positive samples be at least m closer in angular space than negative samples. As a result, it improves alignment for positive pairs while maintaining adequate separation from negatives, leading to more discriminative and well-structured embedding spaces. We demonstrate through Fig. 3.3 that using our angle-based loss can lead to better alignment.

To preserve the structure of the embedding space while enabling a discriminative decision boundary in densely clustered regions, we propose using angular loss. Previous studies on angular loss have focused on single modalities, often adjusting text embeddings. However, this approach falls short for tasks like RIS, where two texts may refer to the same object but are not identical from the text encoder’s perspective. This means that text information must be conditioned on the accompanying image to correctly identify the object within the context. Thus, we leverage text-query-conditioned multimodal embeddings. To model the angular similarity between hidden representations, we apply the arccosine ($\cos^{-1}$) function to the dot product of two normalized representations.

Given a pair of images $\{I_i\}_{i=1}^n$ and their corresponding text queries $\{T_i\}_{i=1}^n$, we construct image-text fused embeddings. These cross-modal embeddings are generated using various fusion strategies. Fusion can occur early in the vision

encoder stages, after feature extraction, or during feature extraction itself. Let F_i denote the resulting cross-modal embedding for the i -th image-text pair.

To measure the similarity between embeddings, we define $\text{cossim}(F_i, F_j)$ as the cosine similarity function for the cross-modal representations of (T_i, I_i) and (T_j, I_j) . Based on this, the angular similarity $\theta_{i,j}$ is expressed as:

$$\theta_{i,j} = \arccos(\text{cossim}(F_i, F_j)), \quad (3.1)$$

where $\theta_{i,j}$ represents the angular distance between F_i and F_j .

However, this form of angular similarity is unsuitable for contrastive learning because it is not a monotonically increasing function. A modified version of the angle-based similarity function is introduced in **Section 3.3**.

3.3 Verb-Centric Multimodal Radial Contrastive Loss

Our goal is to enable Referring Image Segmentation (RIS) models to better understand action-centric verb expressions without requiring complete retraining. To achieve this, we propose the Verb-Centric **Multimodal Radial Contrastive Loss (MRaCL)**, a training paradigm that complements existing objectives and seamlessly integrates into the optimization process. **MRaCL** addresses the limitations of standard image-text pair setups in current RIS datasets by **incorporating an additional loss of verb phrases**. Specifically, embeddings of ground truth image-text pairs are contrasted against cross-modal embeddings derived from isolated verb phrases extracted from in-batch negative texts, as well as their corresponding original full captions.

Unlike traditional cosine similarity-based objectives, **MRaCL** directly ma-



Figure 3.2: **Comparison of Verb Parsing: SpaCy vs. LLaMA.** SpaCy-based rule-based verb parsing is limited, often extracting incomplete or incorrect verbs. LLaMA provides more accurate and context-aware verb extraction, capturing nuanced actions in the captions.

manipulates angular representations by **converting angular distances in radians into logits**. This approach ensures effective gradient flow even in densely clustered multimodal embedding spaces, addressing inherent limitations of cosine similarity such as vanishing gradients and poor separation in anisotropic spaces. By introducing this enhanced angle-based framework, **MRaCL** provides a specialized solution tailored to the unique demands of RIS tasks.

In this context, a “verb phrase” refers not only to single verbs but also to verb-noun pairs and, in some cases, adverbs or prepositional phrases directly associated with the verb to form a complete phrase. By emphasizing such verb phrases, **MRaCL** captures action-centric semantics, including actions, movements, or processes performed by the subject, as well as passive states where the target noun receives the action or occupies a particular state. This verb-centric approach strengthens the model’s ability to ground nuanced, motion-centric language in visual contexts, significantly advancing RIS performance in tasks requiring fine-grained action understanding.

3.3.1 Hard Positive Construction

In order to further encourage our model to focus on verbs, we introduce ‘verb phrases’ as hard positives. To extract verb phrases T_i^{verb} from captions T_i , we utilize Llama3.1-70B with a customized prompt. While traditional part-of-speech (PoS) tagging tools are available, we opted for a large language model (LLM) due to its advantages in handling complex linguistic structures. Specifically, LLMs are better suited for isolating verb phrases that may consist of single verbs, verb-noun combinations, or verb-noun-preposition pairs, depending on the caption. Furthermore, LLMs handle ambiguous cases more effectively, offering robust disambiguation compared to rule-based approaches.

Fig. 3.2 compares verb phrase extraction using Llama3.1-70B and a rule-based method (SpaCy). The rule-based approach struggles to consistently extract complete verb phrases and often misclassifies verbs as nouns, particularly in complex captions. For example, in the caption ‘person doing skateboard tricks’, SpaCy extracts only ‘doing’, while the LLM correctly identifies ‘doing skateboard tricks’ as the complete verb phrase. Rule-based methods also face challenges in accurately associating nouns with their corresponding verbs in long and intricate sentences, even with highly detailed rules. We further validate these advantages through an ablation study in Section 4, demonstrating the effectiveness of our approach in isolating meaningful verb phrases for downstream tasks.

3.3.2 Proposed Method : MRaCL

Now, we propose our **MRaCL**, a simple radial contrastive cross-modal embedding framework. It modifies angular distances in radians into a form suitable

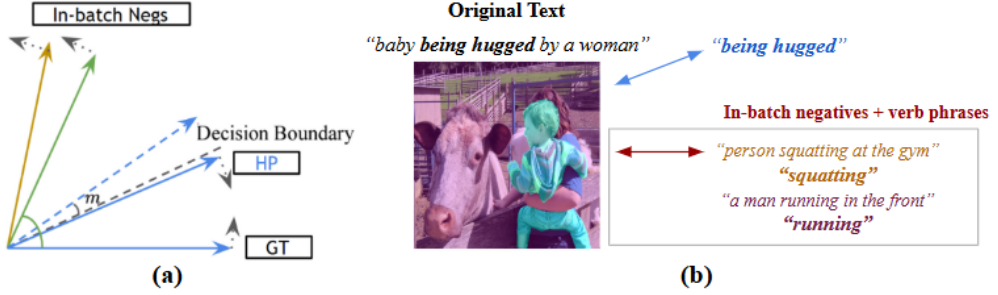


Figure 3.3: (a) **Overview of the MRaCL framework**, which applies a margin penalty to the angle between the anchor and positive. (b) Verb phrase as hard positives. In-batch negatives include unrelated phrases like “person squatting at the gym” with their corresponding verb “squatting”.

for contrastive learning. Specifically, the angular distance is adjusted into a monotonic range aligned with the cross-entropy objective, enabling effective optimization in densely clustered multimodal embedding spaces.

Before utilizing the angle-based function (θ) defined in Equation 3.1, we adjust the range of θ by subtracting a value from $\frac{\pi}{2}$. This adjustment is necessary due to the nature of contrastive learning with a cross-entropy objective, which seeks to increase the similarity of positive pairs while decreasing that of negative pairs. The adjustment shifts the similarity range from $[-1, 1]$ to $[\frac{\pi}{2} - \pi, \frac{\pi}{2} - 0] = [-\frac{\pi}{2}, \frac{\pi}{2}]$:

$$\theta_{i,j} = \frac{\pi}{2} - \cos^{-1}(\text{cossim}(F_i, F'_j)), \quad (3.2)$$

To improve the model’s focus on verb-related semantics, we extract verb-in-action phrases T_i^{verb} from captions T_i using a large language model (LLM). These verb phrases are used as hard positives during contrastive learning. Specifically, we create an additional cross-modal embedding F_i^{verb} using the original image I_i paired with T_i^{verb} . During training, the framework minimizes

the angular distance between F_i and F_i^{verb} , aligning the embeddings of original and verb-focused pairs.

The adjusted loss function, based on angle-based similarity, is defined as:

$$L_{\text{MRaCL}} = -\log \frac{e^{\theta_{i,i^{\text{verb}}}/\tau}}{\sum_{j=1}^N e^{\theta_{i,j}/\tau}}, \quad (3.3)$$

where $\theta_{i,j}$ is the angle-based similarity, and τ is the temperature parameter.

Furthermore, to mitigate the issue of the relatively narrower space (mentioned in Section 3.1), we apply a margin penalty to the angle between the anchor and the positive sample, leveraging its inherent property of angle-based similarity. We simply subtract the angular margin (m) between the anchor (F_i) and the positive pair (F_i^{verb}). The final loss function for MRaCL is defined as:

$$L_{\text{MRaCL}} = -\log \frac{e^{(\theta_{i,i^{\text{verb}}}-m)/\tau}}{e^{(\theta_{i,i^{\text{verb}}}-m)/\tau} + \sum_{j \neq i} e^{\theta_{i,j}/\tau}}, \quad (3.4)$$

where $\theta_{i,j}$ represents the angular similarity between the fused embeddings F_i and F_j , m is the angular margin and τ is the temperature parameter.

Finally, the MRaCL loss is integrated into the overall training objective as:

$$L = L_{\text{ce}} + \alpha \cdot L_{\text{MRaCL}}, \quad (3.5)$$

where L_{ce} is the conventional mask cross-entropy loss, and α is a balancing coefficient. To balance the scale of L_{ce} and L_{MRaCL} , we set $\alpha = 0.1$ by default. The effect of varying α is further analyzed in the ablation study.

Chapter 4

Experiments

4.1 Datasets and metrics

Datasets. We evaluate our approach on four widely used Referring Image Segmentation (RIS) benchmarks: RefCOCO [23], RefCOCO+ [23], G-Ref [36], our M-RefCOCOg and MBench. RefCOCO and RefCOCO+ were created using the ReferItGame, where one player generates a mask and description, and another player identifies the correct mask based on the description within a time limit. This time constraint results in shorter referring expressions, averaging 3 to 4 words. RefCOCO allows positional phrases like "front" or "third from the right," while RefCOCO+ prohibits such references, making it slightly more challenging. In contrast, G-Ref contains longer and more descriptive expressions, averaging 9 to 10 words, often including object relationships, making it the most difficult benchmark. Although we evaluate and report the performance of our method on all three base benchmarks, our primary focus is on the RefCOCOg dataset due to its longer average sentence length and frequent inclusion of verb-in-action

expressions. This makes RefCOCOg an ideal benchmark to assess whether our method performs as expected by effectively leveraging these linguistic cues. Additionally, we evaluate our approach on the M-Ref benchmark, which introduces a motion-centric and static attribute split, as well as the newly proposed MBench, to provide a comprehensive evaluation of our method.

Metrics. We use three metrics for evaluation. First, overall intersection over union (oIoU) measures the ratio of the intersection area to the union area across all test samples. Since oIoU tends to favor larger objects, we also report mean intersection over union (mIoU), which calculates the average IoU between predictions and ground truth. Lastly, we use Precision@ p , which measures the proportion of test samples where the IoU exceeds a specified threshold. The thresholds used are 0.5, 0.7, and 0.9.

4.2 RIS Models

To demonstrate the effectiveness of our data augmentation approach, we experiment with five state-of-the-art RIS models: LAVT [48], CRIS [44], CGFormer [40], and CrossVLT [6]. LAVT employs a late fusion strategy, CRIS and CGFormer utilize early fusion, and CrossVLT adopts a joint fusion approach. This diverse selection of models allows us to evaluate the compatibility and robustness of our framework across different fusion methods. By comparing each model’s performance on RIS tasks with and without incorporating our framework, we aim to show that our approach is effective regardless of the specific visual-text encoder or fusion strategy employed.

4.3 Implementation Details

We cross-validate α within $[0.05, 0.5]$ and choose 0.1 as the default. For m , we perform grid search over $[0, 20]$, and for τ , we search over $[0.01, 0.20]$. Based on these results, we empirically set (m, τ) to $(10, 0.05)$ for all dataset settings. For most of the RIS models, we employ the Swin Transformer Base [33] and BERT [10] as the visual and text backbones, respectively. In the case of CRIS, we use ResNet50 [15] and Transformer [41] backbones. We use the AdamW [47] optimizer with learning rates of $5 \cdot 10^{-5}$ for LAVT, $3 \cdot 10^{-5}$ for CrossVLT, and 10^{-4} for CGFormer and CRIS. The number of training epochs is consistent with the original papers. The batch size is set to 32 for LAVT, 6 for CrossVLT, and 64 for CGFormer and CRIS. Unless otherwise noted, we follow the hyperparameter settings and implementation details from the respective original papers.

4.4 Effectiveness of the Proposed Method

Section 4.4 presents the performance comparison of various RIS models with and without applying our proposed MRaCL, measured in terms of Overall IoU. Across all datasets and fusion strategies, MRaCL consistently improves performance, demonstrating its effectiveness and general applicability regardless of the underlying RIS model. This highlights the robustness of our method in enhancing multimodal embeddings and improving alignment for better object localization.

An interesting observation is the varying degree of improvement across datasets. On average, we observe a more significant performance boost on the more complex RefCOCOg (G-Ref) dataset compared to the simpler RefCOCO

Methods	Encoders		RefCOCO			RefCOCO+			RefCOCOg	
	Visual	Textual	val	testA	testB	val	testA	testB	val(U)	test(U)
<i>Late Fusions</i>										
CRIS [44]	CLIP _{R50}	CLIP	66.68	70.62	59.93	56.94	64.20	46.97	55.91	58.50
CRIS + MRaCL	CLIP _{R50}	CLIP	67.66	71.82	61.06	58.14	65.04	47.84	58.05	59.38
CGFormer [40]	Swin-B	BERT	72.53	75.12	70.09	63.55	68.58	56.05	62.92	64.63
CGFormer+ MRaCL	Swin-B	BERT	73.12	76.05	70.94	64.13	69.28	57.45	64.81	65.12
<i>Early Fusions</i>										
LAVT [48]	Swin-B	BERT	72.73	75.82	68.79	62.14	68.38	55.10	61.24	62.09
LAVT + MRaCL	Swin-B	BERT	73.25	76.12	69.03	62.52	69.45	56.12	63.22	63.92
<i>Joint Fusions</i>										
CrossVLT [6]	Swin-B	BERT	73.44	76.16	70.15	63.60	69.10	55.23	62.68	63.75
CrossVLT + MRaCL	Swin-B	BERT	74.15	76.98	70.73	64.52	70.06	56.61	64.35	65.03

Table 4.1: Performance comparison (in oIoU) with and without MRaCL, across different fusion methods on RefCOCO, RefCOCO+, and RefCOCOg datasets. The highest scores in each category are highlighted in bold.

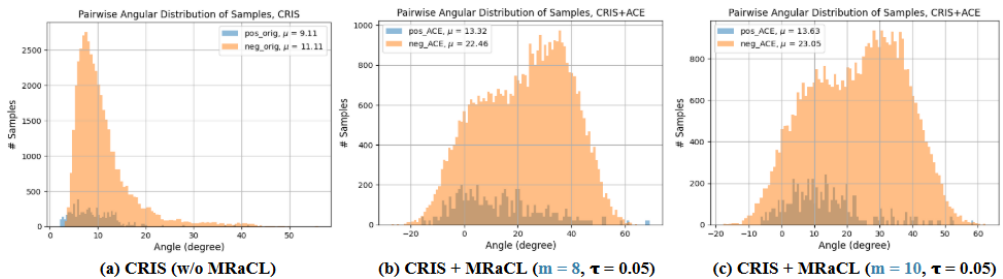


Figure 4.1: **Histogram of angular distributions between cross-modal representations.** (a) CRIS without MRaCL shows densely clustered embeddings. (b) and (c) illustrate MRaCL with $m = 8$ and $m = 10$, where larger margins pull positive samples closer (smaller positive angle mean) and increase separation from negatives, enhancing alignment and diversity..

and RefCOCO+ datasets. Notably, RefCOCOg contains longer queries on average and includes a higher proportion of verb-in-action expressions (approximately 45%) compared to RefCOCO and RefCOCO+ (around 10%). This richer linguistic complexity and emphasis on action-centric semantics make RefCOCOg more challenging, which allows MRaCL to showcase its strength in capturing nuanced, verb-driven relationships and grounding them in visual contexts.

In contrast, the simpler datasets, RefCOCO and RefCOCO+, exhibit smaller performance gains. These datasets primarily consist of shorter, noun-centric queries that focus more on identifying objects through their attributes and spatial relations rather than through dynamic or action-based descriptions. As a result, the role of verb understanding is relatively diminished in these datasets. Nevertheless, the observed improvements, though modest, demonstrate that MRaCL is still effective even in scenarios where the importance of verbs is limited. Importantly, this indicates that our method not only enhances performance but does so without negatively affecting the simpler cases.

Overall, these results confirm that MRaCL is particularly beneficial for tasks involving complex, action-oriented language while maintaining consistent gains across simpler datasets. Its ability to generalize across diverse scenarios highlights its versatility and importance for RIS tasks.

4.5 Effectiveness of MRaCL on Action-oriented query understanding

Section 4.5 presents the performance comparison of RIS models with and without MRaCL on both the M-RefCOCOg split and MBench. The results demonstrate that our method consistently improves performance across both static and motion splits. However, the improvement is notably more significant in the motion-centric split, highlighting the effectiveness of MRaCL in handling action-oriented expressions.

For the M-RefCOCOg benchmark (Section 4.5 (a)), which divides examples into motion and static splits, MRaCL achieves higher performance improve-

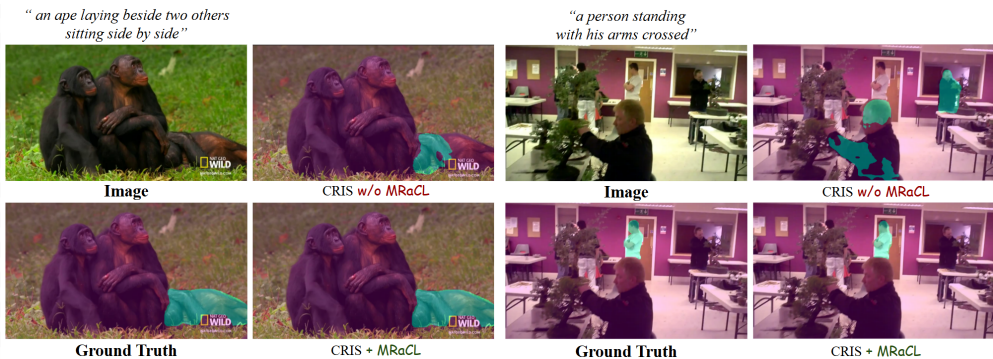


Figure 4.2: Visualization of predictions from the MBench test set.

ments in the motion split compared to the static split. This indicates that our approach is particularly effective in scenarios where the referred object must be disambiguated based on motion-related cues rather than static attributes such as color, texture, or position. For instance, in the motion split, the improvement of mIoU with CGFormer is +4.36%, compared to +0.26% in the static split. Similarly, LAVT and CRIS also exhibit significant gains in the motion split, with improvements of +5.92% and +5.29%, respectively, further highlighting MRaCL’s strong impact on motion-centric scenarios.

(a) *M-RefCOCOg* Motion and Static Split

Options	mIoU		oIoU	
	static	motion	static	motion
CRIS	70.36	52.82	65.28	48.37
CRIS + MRaCL	71.64	58.11	67.08	52.85
LAVT	72.59	55.05	70.63	49.78
LAVT + MRaCL	73.23	60.97	71.75	57.01
CGFormer	75.86	58.22	74.95	52.27
CGFormer + MRaCL	76.12	62.58	75.86	56.64

(b) *MBench* Evaluation Result

Options	mIoU (MBench)	oIoU (MBench)
CRIS	30.47	25.26
CRIS + MRaCL	35.01	27.42
LAVT	33.32	26.54
LAVT + MRaCL	38.29	31.05
CGFormer	34.59	28.21
CGFormer + MRaCL	39.12	32.58

Table 4.2: Performance comparison of models on M-RefCOCOg and MBench.

On MBench (Section 4.5 (b)), a more challenging benchmark designed for action-centric scenarios, MRaCL shows consistent improvements across all models. MBench contains highly visually ambiguous examples, with multiple similar-looking objects and diverse actions performed by referents. Despite these chal-

lenges, MRaCL leads to significant gains, particularly in the motion split, where it enhances the disambiguation of objects based on dynamic action cues. For instance, CGFormer with MRaCL improves mIoU by +4.53% and oIoU by +4.37%, further demonstrating the robustness of our approach in complex, motion-heavy settings.

Qualitative results in Fig. 4.2 further illustrate the effectiveness of MRaCL. In the first example, the query “an ape laying beside two others sitting side by side” requires distinguishing a specific object based on its relative position and interaction. Without MRaCL, CRIS fails to localize the target ape correctly, instead highlighting the incorrect regions. In contrast, CRIS with MRaCL successfully identifies the correct target, grounding the motion-related semantics described in the query. Similarly, in the second example, “a person standing with his arms crossed,” MRaCL enables better focus on the specified action (“arms crossed”), improving segmentation accuracy even in a visually cluttered environment. These examples highlight how MRaCL effectively grounds dynamic and motion-centric language into visual contexts, addressing the challenges posed by action-oriented benchmarks like MBench.

4.6 Ablation Study

4.6.1 Hyperparameter Selection

We conduct an ablation study to analyze the contributions of individual hyperparameters in our MRaCL framework. All experiments are performed with CRIS [44] on the UMD validation set of G-Ref. The study focuses on three key hyperparameters: margin, temperature (τ), and metric loss weight (α).

Choices	val(umd)		Prec(val)		
	mIoU	oIoU	0.5	0.7	0.9
Margin					
8	59.93	57.53	69.24	56.61	15.17
10 (our)	60.31	58.05	69.38	57.45	15.56
12	59.07	56.25	68.56	55.94	14.45
15	60.13	57.73	68.90	56.59	15.14
20	60.08	57.12	68.67	56.54	15.18
Temperature (Tau)					
0.01	59.86	57.01	68.90	56.37	15.12
0.05 (our)	60.31	58.05	69.38	57.45	15.56
0.10	60.22	57.36	69.12	57.26	15.53
0.15	60.05	57.10	68.85	56.80	15.30
0.20	59.74	56.75	68.62	56.13	14.92
Metric Loss Weight					
0.05	60.17	57.25	69.12	56.33	15.93
0.10 (our)	60.31	58.05	69.38	57.45	15.56
0.15	60.13	57.68	68.63	57.38	15.53
0.10	59.64	56.13	68.51	55.47	15.23
0.25	59.43	55.54	68.35	55.13	14.01

Table 4.3: RefCOCOg ablation result on hyperparameter selection.

Margin. We fix the metric loss weight ($\alpha = 0.1$) and temperature ($\tau = 0.05$) while varying the margin values. As shown in Table 4.3, the performance is sensitive to the margin setting. Margins that are too small fail to enforce sufficient separation between positive and negative samples, while excessively large margins lead to collapse and reduced performance. The best performance is achieved around $10 \leq m \leq 15$, with a margin of 10 yielding the highest mIoU (60.98%) and precision scores. This suggests that a balanced margin is crucial for effective contrastive alignment.

Temperature (τ). Fixing the metric loss weight ($\alpha = 0.1$) and margin ($m = 10$), we evaluate the effect of temperature. Lower temperatures ($\tau \leq 0.05$) provide sharper gradients, leading to better alignment, while higher values dilute the effect of the angular similarity. A temperature of 0.05 delivers the

best results with an mIoU of 60.98%, confirming that fine-tuning τ is essential for optimizing performance.

Metric Loss Weight (α). The metric loss weight α determines the relative contribution of L_{MRaCL} compared to the cross-entropy loss (L_{ce}). Fixing the margin ($m = 10$) and temperature ($\tau = 0.05$), we vary α . As noted in Section 3.3.1, $\alpha = 0.1$ was chosen as the default since it balances the scale of L_{MRaCL} (approximately one-tenth that of L_{ce}). Experimental results validate this choice, with $\alpha = 0.1$ achieving the highest mIoU (60.98%). Larger weights (e.g., $\alpha = 0.2$) tend to overemphasize the metric loss, leading to slight performance drops.

Option	mIoU	oIoU	Prec(val)		
	val(umd)	val(umd)	0.5	0.7	0.9
CRIS	59.17	55.91	67.95	55.45	15.27
CRIS + L2	58.99	55.17	67.85	55.09	14.87
CRIS + SimCSE	59.77	56.09	68.15	55.45	14.42
CRIS + MRaCL (ours)	60.98	58.05	69.38	57.45	15.56

Table 4.4: Comparison to other contrastive loss functions.

4.6.2 Comparison to Other Contrastive Loss Functions

Table 4.4 presents a comparison of MRaCL (ours) with other commonly used contrastive loss functions, including L2 and SimCSE, on the UMD validation set of RefCOCOg. For each loss, a grid search was performed to determine the best hyperparameter settings, and the corresponding results are reported.

The results clearly demonstrate the superiority of our proposed method. While L2 achieves subpar performance (59.17% mIoU) compared to CRIS alone, a closer analysis of the distance matrix reveals that the embedding space collapses, resulting in overly similar embeddings across all inputs. This highlights

the sensitivity of L2 to hyperparameter tuning and its inability to generalize effectively in this setting.

SimCSE performs better than L2 (59.77% mIoU) but fails to match the performance of MRaCL. This is likely due to SimCSE’s reliance on standard cosine similarity, which struggles to handle the anisotropic nature of the multimodal embedding space in our task. In contrast, MRaCL, with its angle-based similarity and margin adjustment, effectively addresses these challenges, achieving the best performance across all metrics. These findings emphasize the robustness and adaptability of MRaCL to the multimodal embedding space.

Chapter 5

Conclusion

In this work, we addressed the limitations of existing Referring Image Segmentation (RIS) models in understanding motion-centric expressions, which are critical for resolving visual ambiguities in multimodal tasks. By introducing the **Multimodal Radial Contrastive Loss (MRaCL)**, we leveraged verb phrases as hard positives and employed an adjusted angular margin contrastive loss to overcome the challenges posed by densely clustered multimodal embedding spaces. Our method demonstrated superior performance across diverse benchmarks, including motion-centric splits and our novel action-centric evaluation benchmark, outperforming conventional cosine-based approaches. These results highlight the effectiveness of MRaCL in enhancing both the linguistic and visual comprehension of RIS models, providing a robust framework for tackling complex, action-oriented scenarios. Future work may explore extending this approach to other multimodal tasks requiring fine-grained verb and action understanding.

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초 록

Referring Image Segmentation (RIS)은 여러 가지 모달리티 정보에 대한 복합적인 이해가 필요하며, 특히 동작 중심 표현(action-centric expressions)을 해석할 때 높은 정확도를 요구한다. 최근 RIS 데이터셋은 이미지와 텍스트의 복잡도가 증가하며, 객체 간 상호작용과 세분화된 텍스트-이미지 관계 이해 과정에서 동작 표현을 활용하는 것의 중요도가 점차 높아지고 있다. 특히, 동사는 텍스트 내에서 객체를 구분하는 핵심 단서로 작용하며, 복잡한 데이터셋일수록 그 중요성이 부각된다. 하지만, 기존 모델은 종종 동사와 이와 관련된 의미를 효과적으로 활용하지 못하여, 동작 지향적 질의에 대한 성능이 저하되는 문제가 있다. 이를 해결하기 위해, 우리는 새로운 학습 목표인 Verb-Centric Multimodal Radial Contrastive Loss (MRaCL)을 제안한다. MRaCL은 각도 표현(angular representations)을 직접 조작하여 라디안 단위의 각도 거리를 로그로 변환함으로써, 더 효과적인 그래디언트 흐름을 제공하고 여러 모달리티 정보가 밀집된 공간에서 임베딩 간 분리를 용이하게 한다. 이를 통해 동작 중심 언어와 해당 시각적 단서를 보다 효과적으로 정렬한다. 우리의 방법은 동작 중심 벤치마크에서 RIS 성능을 크게 향상시키며, 역동적이고 섬세한 언어-시각 관계를 이해하는 모델의 능력을 한층 발전시킨다.

주요어: 멀티모달, 동작 중심 이해, 참조 이미지 분할, 대조 손실, 각도 기반 손실

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